

MODEL QUESTION PAPER

Revision2010

TED (10)4095

Fifth Semester Diploma Examination in Polymer Technology

Maximum Marks ;100

FIBERS AND COMPOSITES

Time 3Hours

PART –A (10marks)

- 1 . Answer the following questions in one or two sentences. Each question carries 2 marks.
- a. Define Elastomer
 - b. What is Xanthation of Rayon fiber
 - c Name two Thermosetting plastics
 - d . What is a “ pre- form”
 - e . Expand the term PHBCM (5x2=10)

PART-B

(30 marks)

11. Answer any five full questions from the following Each question carries 6 marks .
- a . Illustrate the classification of Fibers
 - b Describe the processing of Asbestos fiber
 - c List the different grades of Glass
 - d Define the terms Tenacity ,Tex ,Denier and Crystallinity
 - e Illustrate the reaction of Phenol and Formaldehyde (novolac)
 - f Describe the changes occur due to the addition of, filler, and metallic powder in a polymeric composite
 - g Write a formulation for the production of water tanks and justify the selection ingredients

(5x6=30)

PART-C

(60marks)

Answer one full question from each module. Each question carries 15 marks

MODULE -I

- III (a) Distinguish , fiber. Fiber forming plastics and elastomers (6)
- (b) Describe the production of wool (5)

(c) List any four properties of fiber (4)

OR

IV(a) Describe the collection, cultivation ,and application of Cotton fiber (6)

b Describe the production and properties of Sisal and Coir fiber (6)

c What is 'retting ' of fiber (3)

MODULE-II

V (a) Explain the production application and properties of Viscose rayon (7)

(b) Explain the production and applications of Carbon and Aramid fibers (8)

OR

VI (a) Explain the properties and structure of nylon 6 and Melt spinning of nylon 6 (8)

b) Describe the manufacturing steps of Unsaturated Poly Ester resin , Give its important properties (7)

MODULE-III

VII (a) Explain the various additives in FRP technology (8)

(b) Describe the manufacture of Melamine Formaldehyde resin .Give its properties and applications (7)

OR

VIII (a) Explain the manufacture of Polyurethane . Give its applications (8),

(b) Define Thermoset plastics . Explain the production Epoxy resins (7)

MODULE-IV

IX (a) Explain the process of Hand- lay- up and Spray- lay -up techniques (8)

(b) Describe RRIM process (7)

OR

X (a) Write a formulation for Bath-tub and describe its production process (6)

b) Distinguish SMC & DMC (3)

c) Describe the process of Pultrusion (6)

[illegible]